

MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

Texts

Primary text

- **LiZh** — Li & Zhao, *Introduction to Abstract Algebra*

Supplementary reading

- **Childs** — Childs, *A Concrete Introduction to Higher Algebra*

Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*

Schedule

Date	Topic	Reading
Thu 20 Aug	Lecture — Course launch. The integers: division algorithm, gcd, Bezout; congruence and the ring $\mathbb{Z}_n$	Childs Ch. 3, 5 & 6.
Tue 25 Aug	Lecture — Binary operations; Cayley tables; $\mathbb{Z}_n$ under addition and multiplication	LiZh §1.1
Thu 27 Aug	Lecture — Isomorphic binary structures	LiZh §1.2
Tue 1 Sep	Lecture — Groups: axioms and first properties. $\mathbb{Z}_n$, the unit group $(\mathbb{Z}_n)^*$, and the symmetries of a square	LiZh §1.3
Thu 3 Sep	Lecture — Subgroups; subgroup criteria; the subgroups of D_n and of $\mathbb{Z}_n$	LiZh §1.4
Tue 8 Sep	Q&A (Due: Set 1)	
Thu 10 Sep	Lecture — Cyclic groups; the order of an element; $\mathbb{Z}_n$, $(\mathbb{Z}_p)^*$ and primitive roots	LiZh §1.5
Tue 15 Sep	Lecture — Generating sets; D_n as the group generated by a rotation and a reflection	LiZh §1.6
Thu 17 Sep	Lecture — Permutation groups; cycle notation; Cayley's theorem	LiZh §2.1
Tue 22 Sep	Lecture — Alternating groups; the sign of a permutation; A_4 and the tetrahedron	LiZh §2.2
Thu 24 Sep	Lecture — Plane isometries and finite symmetry groups: every one is cyclic or dihedral	Instructor notes
Tue 29 Sep	Q&A (Due: Set 2)	
Thu 1 Oct	Exam — Midterm Exam 1 — Sets 1 and 2	
Tue 6 Oct	Lecture — Cosets; Lagrange's theorem; index	LiZh §3.1
Thu 8 Oct	Lecture — Homomorphisms: kernel, image, and first properties	LiZh §3.3
Thu 15 Oct	Lecture — Normal subgroups; quotient groups; $\mathbb{Z}/n\mathbb{Z}$ built properly	LiZh §3.4
Tue 20 Oct	Lecture — The isomorphism theorems; groups of small order	LiZh §3.4

Date	Topic	Reading
Thu 22 Oct	Lecture — Rings and subrings; $\mathbb{Z}_n$ as a ring; units and zero divisors	LiZh §4.1
Tue 27 Oct	Q&A (Due: Set 3)	
Thu 29 Oct	Lecture — Integral domains and fields; $\mathbb{Z}_n$ is a field exactly when n is prime; characteristic. Euler's and Fermat's theorems via $(\mathbb{Z}_n)^*$	LiZh §4.2; LiZh §4.6
Tue 3 Nov	Lecture — Ring homomorphisms, ideals and quotient rings; the isomorphism theorems for rings	LiZh §4.5
Thu 5 Nov	Lecture — Ideal theory: principal, prime and maximal ideals	LiZh §4.7
Tue 10 Nov	Lecture — Irreducibles and primes; unique factorization domains; principal ideal domains	LiZh §5.1-5.2
Thu 12 Nov	Lecture — Euclidean domains; the Euclidean algorithm in $\mathbb{Z}$ and in $k[x]$	LiZh §5.3
Tue 17 Nov	Lecture — Multiplicative norms; the Gaussian integers $\mathbb{Z}[i]$; Fermat's Two Squares theorem	LiZh §5.5; Conrad (statements and examples; proofs optional)
Thu 19 Nov	Q&A (Due: Set 4)	
Tue 24 Nov	Lecture — Sums of three and four squares; $\mathbb{Z}[\sqrt{-5}]$ and the failure of unique factorization — a closing survey	Instructor notes
Tue 1 Dec	<i>Review</i> — Review — the exam pool	
Thu 3 Dec	Exam — Midterm Exam 2 — Sets 3 and 4	